Joshua C. Macdonald

Computational Scientist

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Research Profile

Computational scientist whose work determines what actions to take, what experiments to run, and what measurements are worth collecting in systems where interventions are costly and uncertainty is unavoidable. I develop methods, scientific AI/ML, and research software for partially observed systems across earth, environmental, and health sciences — integrating generative modeling (mechanistic, statistical, and hybrid), Bayesian inference, uncertainty quantification, and scientific computing to build defensible decision support. Current application areas include WHO-facing diphtheria outbreak-response vaccination analyses, respiratory-virus scenario modeling, surveillance design, marine and terrestrial ecology, and cultural transmission dynamics.

Education

PhD **University of Louisiana at Lafayette**, Mathematics — Lafayette, Louisiana, USA, 2017 – 2022

- Advisor: Hayriye Gulbudak
- Dissertation: *Population Dynamics Under Disturbance in Ecology and Epidemiology*

MS **University of Louisiana at Lafayette**, Mathematics — Lafayette, Louisiana, USA, 2017 – 2020

BA **University of North Carolina Asheville**, Mathematics — Asheville, North Carolina, USA, 2014 – 2016

BA **University of North Carolina Greensboro**, Archaeology — Greensboro, North Carolina, USA, 2008 – 2013

Appointments

Johns Hopkins Bloomberg School of Public Health, Assistant Scientist — Baltimore, Maryland, USA, 2026 – present

International Vaccine Access Center, Department of International Health (Global Disease Epidemiology and Control). Mentor: Shaun Truelove. Research focus: infectious disease modeling and inference, scientific computing, decision support systems, and research software engineering.

Johns Hopkins Bloomberg School of Public Health, Postdoctoral Scholar — Baltimore, Maryland, USA, 2024 – 2026

International Vaccine Access Center, Department of International Health (Global Disease Epidemiology and Control). Mentor: Shaun Truelove. Research focus: infectious disease modeling and inference, scientific computing, decision support systems, and research software engineering.

Tel Aviv University, Zuckerman STEM Leadership Postdoctoral Fellow — Tel Aviv, Israel, 2022 – 2024

School of Zoology. Mentor: Yoav Ram. Research drew directly on archaeological training in the study of material culture diversity and cultural transmission, applying Bayesian dimensionality reduction and multivariate data analysis to cultural datasets.

Computational Expertise

Modeling: Generative models (ODE/PDE/stochastic/hybrid); compartmental and agent-based models; scientific AI/ML (physics-embedded surrogates, lawful learning); multivariate data analysis and dimensionality reduction; stability, bifurcation, and sensitivity analysis; parameter identifiability and inverse problems

Inference: Bayesian hierarchical models; variational and simulation-based inference; data assimilation; uncertainty quantification; model calibration and validation; proper scoring rules and forecast evaluation

Computing: Python, Julia, C++, Stan, R, MATLAB. Scientific computing; high-performance numerical solvers; operator splitting; data pipelines and exploratory analysis; reproducible workflows (Git, CI/CD, containerization)

Published Papers

An immuno-epidemiological model of foot-and-mouth disease in African buffalo with carrier transmission 2026
S. Atkins, H. Gulbudak, S. Welker, H. Smith, ***J.C. Macdonald***, B.R. Beechler, A.E. Jolles
(Bulletin of Mathematical Biology)

Within-host viral growth and immune response rates predict FMDV transmission dynamics for African buffalo 2024
J.C. Macdonald, H. Gulbudak, B. Beechler, E. Gorsich, S. Gubbins, E. Perez, A. Jolles
10.1086/730703 (The American Naturalist, 204(2), 133–146)

Cultural transmission, networks, and clusters among Austronesian-speaking peoples 2024
J.C. Macdonald, J. Blanco-Portillo, M.W. Feldman, Y. Ram
10.1017/ehs.2024.45 (Evolutionary Human Sciences, 6, e45)

Forward hysteresis and Hopf bifurcation in an NPZD model with application to harmful algal blooms 2023
J.C. Macdonald, H. Gulbudak
10.1007/s00285-023-01969-7 (Journal of Mathematical Biology, 87, 45)

First and second wave outbreak size with reactive quarantine measures: a COVID-19 model applied to an epidemic in China 2022
C.J. Browne, H. Gulbudak, ***J.C. Macdonald***
10.1016/j.jtbi.2021.110919 (Journal of Theoretical Biology, 538, 110919)

Modeling COVID-19 outbreaks in the United States with distinct testing, lockdown speed, and fatigue rates 2021
J.C. Macdonald, C.J. Browne, H. Gulbudak
10.1098/rsos.210227 (Royal Society Open Science, 8, 210227)

Papers Under Review

Multi-Model Insights into Racial and Ethnic Heterogeneities in Early COVID-19 Transmission and Severity in the U.S. 2026
S.J. Bents, S.L. Larsen, S. Jung, E. Howerton, et al., including ***J.C. Macdonald***
(In revision, American Journal of Epidemiology)

Preprints

Bayesian estimation of the number of significant principal components for cultural data 2024
J.C. Macdonald, J. Blanco-Portillo, M.W. Feldman, Y. Ram
10.48550/arXiv.2409.12129

Research Support – Active

USDA-NIFA AFRI US–UK Collaboration (Award #1032897) 2025 – 2028
Quiescence to emergence: Does nairovirus diversity loss increase disease transmission, severity and the risk of emergence? US PIs: Brianna Beechler, Anna Jolles, Cameron Browne, and Hayriye Gulbudak; linked UKRI award BB/Z516934/1 with Emma Thomson, Teresa Lambe, Roman Biek, and Georgina Limon-Vega. Collaborator (40% FTE). Research role: mathematical model parameterization, data analysis, and advising on mathematical and statistical methods.

USDA-ARS Cooperative Agreement (3022-32000-021-018-S) 2025 – present
Investigating the Epidemiology of CCHFV in an Endemic Setting: Uganda. PIs: Bree Beechler and Anna Jolles (Oregon State University). Collaborator. Research role: mathematical modeling of cross-species CCHFV transmission dynamics, surveillance design optimization, and advising on mathematical and statistical methods for decision support.

CDC-RFA-FT-23-0069 (NU38FT000012) 2023 – present
Atlantic Coast Center for Infectious Disease Dynamics and Analytics (ACCIDDA). Directors: Justin Lessler, Kimberly Powers, and Shaun Truelove. 60% FTE. Insight Net center for infectious disease forecasting and outbreak analytics. Research role: operational modeling, forecasting pipeline development, and computational infrastructure.

NSF EEID / UKRI UK–US Collaboration Grant ($4.05M / 5 years) 2022 – 2027
Multi-scale infection dynamics from cells to landscapes: FMD in African buffalo. PIs: Brianna Beechler, Roman Biek, Hayriye Gulbudak, Erin Gorsich, Simon Gubbins, Anna Jolles, and Jan Medlock. Senior Personnel (from 2024). Research role: mathematical model development and parameterization of within-host viral and immune dynamics.

Research Support – Completed

Zuckerman Postdoctoral Scholarship ($120K total) 2022 – 2024
Co-evolution of Cultural and Genetic Traits. Principal Investigator. Competitive postdoctoral fellowship; authored and defended research proposal.

Research Support – Pending

National One Health Mission India ($1M / 3 years) In review
Development of an Integrated One Health Disease Modelling and Digital Health Program in Human, Livestock, and Wildlife. Co-Investigator. Research role: mathematical model parameterization and data analysis; mechanistic modeling for HPAI and anthrax in India; stakeholder training in modeling; and supervision of postdoctoral and doctoral trainees.

Operational Modeling and Policy Support

Diphtheria Outbreak-Response Vaccination Modeling 2026 – present
Johns Hopkins IVAC, in collaboration with WHO. Lead modeler for multi-country, district-level transmission analyses comparing outbreak-response vaccination strategies across diverse diphtheria risk settings. Developed to inform evidence review and SAGE deliberations on diphtheria outbreak-response vaccination policy. Implemented with op_system, FlepiMoP2, JAX, and Diffrax.

COVID-19 Scenario Modeling Hub 2025 – present
ACCIDDA. Led the Round 20 (2026–27) submission, including rebuilding and migrating the legacy model to op_system and FlepiMoP2, model development and calibration, scenario production, quality assurance, and hub submission. Previously contributed data wrangling and data assimilation to the Hub's health-heterogeneities scientific round.

- COVID-19 Scenario Modeling Hub
- github.com/midas-network/covid19-scenario-modeling-hub

Flu Scenario Modeling Hub 2024 – present

ACCIDDA. Lead Model Developer across the 2024/25, 2025/26, and 2026/27 seasons. Set up the ACCIDDA flu model and operational pipeline (2024/25). Extensive work with external modeling packages drove contributions that led to FlepiMoP2 (2025/26). Supervising an undergraduate student in a scientific overhaul of the flu model for the 2026/27 round.

- github.com/midas-network/flu-scenario-modeling-hub

Hib Vaccination Modeling (Navajo Nation) 2026 – present

ACCIDDA. Technical supervisor managing PhD students in implementation of an age- and immune-status-structured Hib model for the Navajo Nation to evaluate the impact of long-running vaccination programs.

Research Software – Lead Developer

OP System (Operator-Partitioned System) 2026 – present

ACCIDDA. Released compiler for structured model specifications, producing Array API right-hand sides, reaction and observable metadata, forcing schedules and checked independent-block contracts. NumPy, JAX and CPU PyTorch qualification; independent-equation fixtures complement structural validation.

- github.com/ACCIDDA/op_system

pp-eigentest 2026 – present

Ongoing research on posterior-predictive signal-rank selection for incomplete datasets, building on a 2024 preprint. Calibration and evaluation studies are in progress; methods paper in preparation and companion software private and pre-release.

- Project summary

op_infer (Inference and Calibration) 2026 – present

Working statistical layer of the op stack, connecting Bayesian fitting through established inference libraries, simulation-based calibration and replicated study-design checks. Private, pre-alpha and unreleased; broader interfaces and application-specific validation remain in development.

- Project summary

Trade Study (Python) 2026 – present

Released framework for replicated simulation studies, paired comparisons, Pareto analysis and preference-based decisions. Hierarchical phases, proper scores, fixed/adaptive replication, sensitivity and surrogate integrations, and provenance-checked result bundles. Statistical guarantees depend on declared assumptions and application-specific validation.

- github.com/jcm-sci/trade-study
- pypi.org/project/trade-study/
- doi:10.5281/zenodo.19599838

OP Engine (Operator-Partitioned Engine) 2025 – present

ACCIDDA. Released numerical core with deterministic/stochastic solvers, independent NumPy SSA banks, named observations and adaptive-schedule replay. Method-specific NumPy, JAX and CPU PyTorch qualification; replay derivatives are conditional on the recorded mesh and GPU paths need separate hardware evidence.

- github.com/ACCIDDA/op_engine

vbpca-py 2024 – present

Variational Bayesian PCA for incomplete data with native per-entry missingness handling, full posterior uncertainty quantification, automatic component pruning, and built-in model selection. Missing-aware preprocessing pipeline and C++-accelerated kernels; scikit-learn-compatible API. Applied to genetic, cultural, and ecological datasets.

- pypi.org/project/vbpca-py/
- doi:10.5281/zenodo.19389250

Research Software – Collaborative Infrastructure

FlepiMoP2 (Flexible Pipeline for Interchangeable Model Processing) 2025 – present

ACCIDDA. Core contributor to the configuration-driven orchestration engine for infectious disease forecasting and scenario analysis. Plugin architecture decouples model specification, numerical integration, and output persistence; led migration of ACCIDDA's COVID-19 model for the Round 20 Scenario Modeling Hub submission.

- Documentation

TabularEpi 2025 – 2025

UNC / ACCIDDA. Tabular epidemic simulator supporting mixed deterministic-stochastic models. Conducted systematic performance benchmarking that identified bottlenecks, informing the design of the next-generation pipeline.

FlepiMoP (Flexible Epidemic Modeling Pipeline) 2024 – 2025

Hopkins IDD / ACCIDDA. Developed simulation and statistical inference components for operational influenza modeling; conducted benchmarking that identified 5–20x performance improvements, informing the next-generation pipeline architecture.

Invited Talks and Seminars

Cross-Scale Feedback Motifs: Structure-Preserving Models and Computational Tools for Complex Systems Mar 2026

Biological Oceanography Seminar, Woods Hole Oceanographic Institution, Woods Hole, Massachusetts

Decision-Support Modeling for One Health Pathogens: Using Mechanistic Models for Surveillance and Forecast Design Mar 2026

Differential Equations Seminar, University of Maryland Baltimore County, Baltimore, Maryland

Modeling long-term viral infections 2025

Mathematics Seminar, Shippensburg University, Shippensburg, Pennsylvania

Scenario and forecast modeling for respiratory viruses 2025

Navajo Nation Epidemiology Center Data Summit, Window Rock, Arizona

Impacts of host behavior and viral life history on infectious disease transmission 2024

Department of Mathematics, Virginia Tech; Infectious Disease Dynamics Group, Johns Hopkins University

Within-host viral life histories predict FMDV transmission dynamics for African buffalo 2024

Hawlena Lab, Ben-Gurion University, Midreshet Ben Gurion, Israel

Modeling process-based effects of harmful algal blooms 2023 – 2024

University of Essex (2024); Tel Aviv University (2023); Israel Oceanographic and Limnological Research, Haifa (2023); Inter-University Institute for Marine Sciences, Eilat (2023)

Patterns of cultural variation in the Austronesian expansion 2023
Fortunato Lab, University of Oxford (virtual)

Consilience in FMDV ecology 2022
Gresham Lab, New York University; Obolski Lab, Tel Aviv University

Selected Conference Presentations

Decision-Support Modeling for One Health Pathogens: Using Mechanistic Models for Surveillance and Forecast Design 2026
Society for Mathematical Biology Annual Meeting, Graz, Austria; mini-symposium talk

Recovering Ecological Geometry: A Trait- and Depth-Structured IPDE Model of Plankton Dynamics 2025
Society for Mathematical Biology Annual Meeting, Edmonton, Alberta, Canada; mini-symposium talk

Biologically Grounded Surrogate-Driven Parameter Inference for Sparsely Observed Dynamical Systems 2025
Symposium on Biomathematics and Ecology Education and Research (BEER), Fairfax, Virginia; contributed talk

Forward hysteresis and Hopf bifurcation in an NPZD model with application to harmful algal blooms 2023
Society for Mathematical Biology Annual Meeting, Columbus, Ohio; mini-symposium talk

Within-host viral growth and immune response rates predict FMDV transmission dynamics for African buffalo 2022 – 2023
SIAM TXLA, Lafayette, Louisiana (2023); SMB Epidemiology-Immunology Midyear Meeting (2022, virtual)

Dengue immunodynamics: antibody-dependent enhancement, cross-reactive immunity, and model–data confrontation 2022 – 2023
SIAM TXLA (2023); Mathematical Perspectives on Immuno-Biology, Bulgaria (2023); Center for Combatting Pandemics, Tel Aviv (2023); European Conference on Theoretical and Mathematical Biology, Heidelberg (2022)

Disease dynamics necessarily operate across scales 2021
SIAM TXLA, South Padre Island, Texas; mini-symposium talk

Modeling COVID-19 outbreaks in the United States with distinct testing, lockdown speed, and fatigue rates 2021
Society for Mathematical Biology Annual Meeting (virtual); contributed talk

Workshop Facilitation

Insight Net Third Annual Meeting, Tools Workshop Presenter — Chapel Hill, North Carolina, USA
Presented FlepiMoP2 and the op_system/op_engine simulation stack at the Insight Net Third Annual Meeting tools workshop, Friday Center, University of North Carolina at Chapel Hill. Apr 2026 – present

Insight Net Second Tech Transfer Workshop, Technical Support — Chapel Hill, North Carolina, USA
ACCIDDA. Technical support for the Insight Net Second Tech Transfer Workshop at UNC Chapel Hill. Assisted participants with Python package installation and environment setup for Epydemics tools. Jan 2026 – present

ACCIDDA, Workshop Facilitator — Baltimore, Maryland, USA
Facilitator for the Applied Modeling in Public Health (AMPH) Workshop, October 15–17, 2025, Johns Hopkins Bloomberg School of Public Health. — Oct 2025 – present

ACCIDDA, Workshop Facilitator — Baltimore, Maryland, USA
Facilitator for the Applied Modeling in Public Health (AMPH) Workshop, August 12–15, 2024, Johns Hopkins Bloomberg School of Public Health. — Aug 2024 – present

Teaching

Johns Hopkins University, Teaching Assistant — Baltimore, Maryland, USA
Teaching assistant for *Topics in Infectious Disease Epidemiology* with Shaun Truelove and Sara Loo, Summer 2025 and Summer 2026. — 2025 – 2026

University of Louisiana at Lafayette, Teaching Assistant and Instructor of Record — Lafayette, Louisiana, USA
Instructor of record for elementary statistics (3 sections), survey of calculus (2 sections), precalculus (2 sections), and college algebra (2 sections). Teaching assistant for calculus, linear algebra, differential equations, and basic statistics. — 2017 – 2020

McDowell County Schools / Memorial University / UNC Asheville, Teaching and Tutoring — 2016 – 2017
Instructor of record for North Carolina Math III / Algebra II (2017). Mathematics and statistics tutoring at Memorial University of Newfoundland (2016–2017) and University of North Carolina Asheville (2016).

Mentoring

Mentoring students and postdoctoral researchers — 2021 – present
- Jolles and Beechler Labs, Oregon State University (2025–present)
- Society for Mathematical Biology (2022–present)
- Association for Women in Mathematics (2023–2025)
- Ram Lab, Tel Aviv University (2022–2024)
- MIDAS Network (2021–2022)
- Gulbudak Group, University of Louisiana at Lafayette (2021–2022)

Undergraduate Research Supervision: Shany Naim — 2024
Tel Aviv University. Supervised undergraduate research project: initial translation of the VBPCApy variational Bayesian PCA codebase from MATLAB to Python.

Community Outreach

Science communication and research dissemination — 2021 – 2022
- University of Louisiana R1 Showcase: selected to present COVID-19 modeling research at the university-wide showcase (2022)
- Ocean View United Methodist Church, Oak Island, NC: community discussion on vaccines and COVID-19, with on-site pharmacist for vaccination (2021)

Professional Service

SMB Diversity, Equity, and Inclusion Committee — 2023 – 2025
Member

Session Chair / Leader — 2021 – 2023
- SMB Early Career Workshop: Applying for postdoctoral funding (2023)
- SMB Epidemiology-Immunology Midyear Meeting: Contributed talks (2022)
- SMB Annual Meeting: CT04-MEPI contributed talks in mathematical epidemiology (2021)

Mini-symposium Co-organizer 2019 – 2023

- SIAM TXLA: Data-driven mathematical and statistical modeling approaches to population biology (2023)
- SMB Annual Meeting: Disease Dynamics Across Scales (2023)
- SMB/ESTMB Joint Meeting: Multi-scale and data-driven modeling approaches in ecology, immunology, and epidemiology (2022)
- SIAM TXLA: Reproducibility, reliability, and robustness: confronting models from across mathematical biology with data (2021)
- SIAM TXLA: Population Dynamics in Ecology and Epidemiology (2019)

Secretary, University of Louisiana AMS Graduate Student Chapter 2021 – 2022

Journal Reviewing

2022–present: *PLOS ONE*, *Journal of Biological Systems*, *Journal of Biological Dynamics*, *Frontiers in Public Health*, *Frontiers in Veterinary Science*, *Computational and Applied Mathematics*, *Mathematical Biosciences*, *Scientific Reports*, *PLOS Computational Biology*, and *BMC Infectious Diseases*.

Honors and Awards

Travel Awards 2019 – 2022
SIAM, Society for Mathematical Biology, and University of Louisiana Graduate Research Association

Rhodes Outstanding Graduate Assistant Award 2021
Department of Mathematics, University of Louisiana at Lafayette. Awarded to two graduate students annually (approximately 50 graduate students total).

Outstanding Contributed Talk Award (Mathematical Epidemiology) 2021
Society for Mathematical Biology Annual Meeting

Three Minute Thesis Competition, Finalist 2019
Louisiana Sea Grant

Professional Affiliations

Society for Mathematical Biology; Society for Industrial and Applied Mathematics; International Society for Bayesian Analysis; United States Research Software Engineer Association; MIDAS Network; Association for Women in Mathematics; Institute of Industrial and Systems Engineers; Society for Health Systems (SHS/IISE); and Healthcare Information and Management Systems Society (HIMSS).